

Cache Datapath Description

Datapath Function

The datapath handles movement of data between CPU, cache memory, and main memory. It follows controller decisions.

Cache Read Address Generation

cache_read_mode selects between normal CPU access and internal cache line operations such as writeback.

Cache Write Data Selection

cache_write_source selects CPU write data during write hits or memory read data during refill.

Memory Address Generation

memory_addr_mode selects writeback addresses using the old cache tag and refill addresses using the requested tag.

Word Counter

The controller supplied word counter selects individual words during refill and writeback transfers.

Data Paths

CPU read: Cache Memory to Datapath to CPU. Writeback: Cache Memory to Datapath to Main Memory. Refill: Main Memory to Datapath to Cache Memory.